

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology & Geophysics

2. Subject Name: Stratigraphy **L-T-P:** 3-0-0 **Credits:** 3

3. Pre-requisites: NA

4. Syllabus and reference books:

Syllabus:

Definition of stratigraphy: Classification of bedding; Basis of stratification; Types and recognition of stratification; Stratification and rock sequence; Walther's Law; Breaks in record; Marine transgression and regression. Criteria of correlation, dual hierarchy of units of stratigraphy, code of stratigraphic nomenclature; the Stratigraphic System.

Precambrian chronostratigraphy, Chronostratigraphy in Indian cratonic Archean greenstone belts and late Archean sedimentary basins, Archean-Proterozoic boundary reassessment, Proterozoic chronostratigraphy and relation to Indian Purana basins.

Phanerozoic formations of India with special reference to Kashmir, Spiti, Kumaon, Assam Himalayas; Kutch and Trichinopoly areas; Coastal Cenozoic formations. The Gondwana Supergroup: distribution and depositional history; Deccan volcanism. The Archean/Proterozoic, Precambrian/Cambrian, Permian/Triassic, and Cretaceous/Tertiary boundaries in India. The Siwalik Supergroup: tectonic setup, sedimentation, and fauna. Rise of the Himalayas and Pleistocene glaciations. Evolution of the Indian subcontinent.

Reference Books:

- 1) M. Ramakrishnan and R. Vaidyanathan, "Geology of India," Geological Society of India, Bangalore, Vol. 1, 2008.
- 2) M. Ramakrishnan and R. Vaidyanathan, "Geology of India," Geological Society of India, Bangalore, Vol. 2, 2010
- 3) Principles of Sedimentology and Stratigraphy, Sam Boggs Jr., Pearson; 5th edition
- 4) Sedimentology and Stratigraphy, Gary Nichols, Wiley-Blackwell.
- 5) Geological Time Scale 2012 Volumes 1 and 2 (Ed Gradstein F, Ogg JG, Schmitz MD, Ogg GM), Elsevier, 2012
- 6) Cratons and Fold Belts of India, by RS Sharma. Springer, 2009.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1	Stratigraphic principles	4
2	Stratigraphic schemes	8
3	Precambrian Stratigraphy of India	12
4	Phanerozoic Stratigraphy of India	12
Total number of hours		36